PHYSICS

1. The velocity v of a particle of mass m experiencing a resistive force in a fluid is given by the expression, $v = \frac{F_0}{\beta}(1 - e^{-\frac{\beta}{m}t})$, where t is time and F_0 is a constant. Based on the principles of dimensional analysis, The dimensional formula for the constant β?

A) $[MLT^{-1}]$ B) $[MT^{-1}]$ C) $[MLT^{-2}]$ D) $[ML^{-1}T^{-1}]$

2. A wheel of moment of inertia 5 x 10^{-3} kgm^2 is making 20 rev/s. The torque required to stop it in 10 secs is,

A) $2\pi \times 10^{-2}\ Nm$ B) $\pi \times 10^{-2}\ Nm$ C) $2\pi \times 10^{2}\ Nm$ D) $4\pi \times 10^{-2}\ Nm$

3. A bob is hanging from the ceiling of a car with a string, and acts as an accelerometer. If θ is the angle the string makes with vertical, acceleration of car is.

A) a = g$\tan\theta$ B) $a = g sin\theta$ C) $a = g\ cot\theta$ D) $a = g\ cos\theta$

4. A person can throw a ball at maximum distance d over a ground. What is the maximum vertical height to which he can throw?

A) d B) d/2 C) d/4 D) 2d

5. Two horizontal capillary tubes, A and B, of equal lengths but having internal radii in the ratio 1 : 2, are connected in series. A viscous fluid flows steadily through them under a constant pressure difference. If the pressure drop across tube B is ΔP, what is the pressure drop across tube A based on Poiseuille's principle?

A) 16ΔP B) 8ΔP C) 4ΔP D) ΔP

6. Aparticle of mass m executes simple harmonic motion (SHM) with amplitude A and angular frequency ω. What is the average kinetic energy of the particle, calculated over one complete period of oscillation?

A) ½ $m\omega^2A^2$ B) ¼ $m\omega^2A^2$ C) $1/\pi\ m\omega^2A^2$ D) $m\omega^2A^2$

7. Two solid, uniform concentric spheres have masses M and 3M, and radii R and 2R respec tively. Applying the laws of gravitation, what is the magnitude of the net gravitational field strength at a radial distance of 1.5R from their common center?

A) $4GM/9R^2$ B) $16GM/9R^2$ C) GM/R^2 D) $2GM/3R^2$

8. An elastic wire of length L, uniform cross-sectional area A, and Young's modulus Y is stretched longitudinally by an applied force F. If the internal atomic structure of the material allows it to be perfectly incompressible under stress, what is its Poisson's ratio σ?

A) 0.25 B) 0.33 C) 0.5 D) 1.0

9. Acapillary tube of radius r is vertically lowered into a liquid of density ρ and surface tension T. Due to capillary action, the liquid rises to an equilibrium height h. Assuming the angle of contact is perfectly 0^0, what is the total stored gravitational potential energy of the liquid column raised inside the capillary?

A) $\pi T^2/\rho g$ B) $2\pi T^2/\rho g$ C) $\pi r^2 T/\rho g$ D) $T^2/\rho g$

10. A small bob of mass m is attached to an inextensible string of length L and whirled in a vertical circle. The velocity at the lowest point is provided just sufficiently so that the tension at the highest point of the circle is zero. What is the magnitude of the centripetal acceleration of the bob at the instant the string is exactly horizontal?

A) g B) 2g C) 3g D) 4g

11. Two identical spherical black bodies, A and B, have radii in the ratio 1:2 and are maintained at absolute temperatures 2T and T respectively. The ratio of the rates of thermal energy radiated by spheres A and B is:

A) 1 : 1 B) 2 : 1 C) 4 : 1 D) 1 : 4

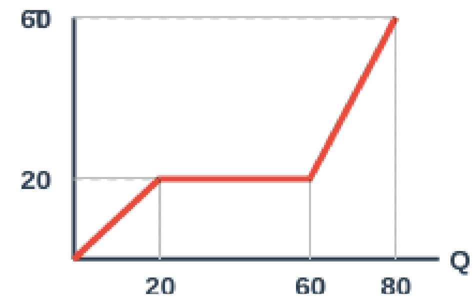
12. A thermometer has wrong calibration it reads the MP of ice −10◦C, it reads 60◦C in place of 50◦C. What is the temp of BP of water on this scale.

A) 150^0C B) 120^0C C) 130^0C D) 140^0C

13. A liquid with a real coefficient of cubical expansion γ is filled in a uniform container. When the system is heated, the apparent level of the liquid with respect to the container remains perfectly unchanged. What must be the coefficient of linear expansion (α) of the material of the container?

A) $\alpha = \gamma$ B) $\alpha = \gamma / 2$ C) $\alpha = \gamma / 3$ D) $\alpha = 3\gamma$

14. The temperature-heat (T-Q) graph for a substance of mass m is shown above. Assuming the substance starts as a solid at T=0, what is the ratio of its specific heat capacity in the solid state to that in the liquid state?

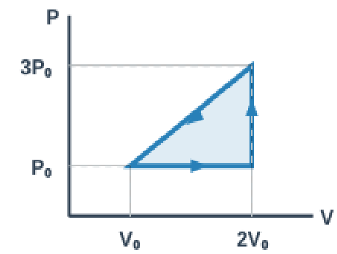


A) 1 : 2
B) 2 : 1
C) 1 : 4
D) 4 : 1

15. An ideal gas is confined in a rigid container. If the absolute temperature of the gas is doubled and its pressure is simultaneously halved by letting some gas escape, what happens to the root-mean-square (rms) speed of the gas molecules?

A) It remains unchanged. B) It becomes half of its original value. C) It increases by a factor of $\sqrt{2}$. D) It doubles.

16. A fixed mass of an ideal gas undergoes a cyclic process as shown in the P-V diagram above. The process proceeds in a clockwise direction. What is the net thermodynamic work done by the gas in one complete cycle?



A) P_0V_0
B) $2P_0V_0$
C) $-P_0V_0$
D) Zero

17. A Carnot engine operates between two thermal reservoirs at temperatures T_1 and T_2 ($T_1 > T_2$) and has an efficiency η. If the identical machine is operated in reverse as an ideal refrigerator between the same two reservoirs, its coefficient of performance (β) is given by:

A) $\beta = \eta / (1 - \eta)$ B) $\beta = (1 - \eta) / \eta$ C) $\beta = 1 / \eta$ D) $\beta = 1 - \eta$

18. If the pressure of an ideal gas is isothermally doubled, what will happen to the speed of sound propagating through that gas?

A) It will double B) It will halve C) It will increase by a factor of $\sqrt{2}$ D) It will remain unchanged

19. An open organ pipe and a closed organ pipe happen to have the exact same fundamental frequency. What is the ratio of the frequency of their first overtones?

A) 1:2 B) 2:3 C) 3:4 D) 1:3

20. Anobserver moves with a speed of v/2 (where v is the speed of sound) directly away from a stationary source emitting frequency f, and directly towards another identical stationary source. What is the beat frequency heard by the observer?

A) 0 B) f/2 C) f D) 3f/2

21. Asymmetric biconvex lens forms a sharp image of a point object with intensity I and focal length f. If the central circular portion of the lens, corresponding to exactly half of its total diameter, is painted black, what happens to the new focal length and image intensity?

A) f and 3I/4 B) f and I/2 C) f/2and I/2 D) 2f and I/4

22. In Young's double-slit experiment, placing a thin glass plate ($\mu = 1.5$) in front of one slit shifts the central maximum by 4 fringe widths. If this glass plate is replaced by a water column ($\mu = 4/3$) of the exact same thickness, by how many fringe widths will the central maximumshift?

A) 2.0 B) 2.67 C) 3.0 D) 4.0

23. To significantly increase the resolving power of an astronomical telescope used to view a distant binary star system, one should:
A) Increase the focal length of the objective lens B) Decrease the focal length of the objective lens
C) Increase the diameter of the objective lens D) Decrease the objective diameter

24. Unpolarized light is incident on a transparent medium of refractive index √3 at the exact polarizing angle (Brewster's angle). What is the angle of refraction inside the medium?
A) 30^0 B) 45^0 C) 60^0 D) 90^0

25. Two thin lenses of focal lengths f_1 and f_2, made of materials with dispersive powers ω_1 and ω_2, are combined to form an achromatic doublet. If the first material has twice the dispersive power of the second ($\omega_1 = 2\omega_2$), what is the relationship between their focal lengths?
A) $f_1 = 2f_2$ B) $f_1 = -2f_2$ C) $f_2 = -2f_1$ D) $f_2 = 2f_1$

26. According to the Standard Model of particle physics, a proton is a baryon composed of which specific combination of quarks?
A) Two up quarks and one down quark (uud) B) One up quark and two down quarks (udd)
C) One up quark, one strange quark, and one down quark (usd) D) Two charm quarks and one bottom quark (ccb)

27. A radioactive sample has a half-life of T. What fraction of the initial active nuclei will DECAY in a time equal to its mean life (τ)?
A) 1/e B) 1 - (1/e) C) 1/2 D) 1 - (1/ln 2)

28. In a photoelectric effect experiment, if the frequency of the incident radiation is exactly doubled (and remains above the threshold frequency), the stopping potential will:
A) Exactly double B) More than double C) Less than double D) Remain the same

29. An electron is accelerated from rest through a potential difference V. If the De-Broglie wavelength of this electron is numerically equal to the minimum wavelength of X-rays produced by an X-ray tube operating at the identical potential V, what is the value of V? (Let m be electron mass, e be charge, c be speed of light)
A) $mc^2/2e$ B) $2mc^2/e$ C) hc/e D) h/mc

30. NAND and NOR gates are known as universal gates. What is the minimum number of two-input NAND gates required to construct an exclusive-OR (XOR) gate?
A) 3 B) 4 C) 5 D) 6

31. The curve of binding energy per nucleon versus mass number (A) shows that stability varies among nuclei. Which of the following nuclei has the maximum binding energy per nucleon?
A) Helium-4 (^{4}He) B) Carbon-12 (^{12}C) C) Iron-56 (^{5}Fe) D) Uranium-235 (^{235}U)

32. In the spectrum of a hydrogen atom, what is the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series?
A) 5/27 B) 4/9 C) 1/3 D) 27/5

33. An electron enters a region of uniform magnetic field with a velocity completely perpendicular to the field lines. Which of the following physical quantities of the electron remains strictly constant during its motion?
A) Velocity B) Momentum C) Kinetic Energy D) Acceleration

34. When a p-n junction diode is heavily reverse-biased, what happens to the thickness of its depletion layer?
A) It decreases significantly. B) It remains completely unaffected.
C) It increases. D) It alternately increases and decreases.

35. Two radioactive materials A and B have decay constants of 10λ and λ, respectively. If initially they have an identical number of active nuclei, after what time will the ratio of the number of active nuclei of A to that of B be precisely 1/e?
A) 1/9λ B) 1/10λ C) 1/11λ D) 9/λ

36. According to Hubble's Law, which forms a cornerstone of the Big Bang theory, the recessional velocity of a distant galaxy:
A) Inversely proportional to the square of its distance from Earth B) Directly proportional to its distance from Earth
C) Independent of its distance from Earth D) Directly proportional to its mass

37. In a Millikan's oil drop experiment setup, a particular oil drop is calculated to have a net charge of -4.8×10^{-19} C. What can be deduced about this specific oil drop?
A) It has lost 3 electrons. B) It has gained 3 protons.
C) It has gained exactly 3 electrons. D) Such a charge is physically impossible due to quantization.

38. Two identical small conducting spheres carry charges of +q and -3q. They attract each other with a force F when separated by a distance d. If they are allowed to touch each other and are then replaced at the same distance d, what is the new force between them?

A) F/3 (attractive) B) F/3 (repulsive) C) F (repulsive) D) 3F (attractive)

39. A hollow conducting sphere of radius R is given a charge +Q. Which of the following correctly describes the electric potential V as a function of distance r from the center of the sphere?

A) V is zero everywhere inside the sphere and decreases as 1/r outside.
B) V is constant inside the sphere and decreases as $1/r^2$ outside.
C) V is constant inside the sphere and decreases as 1/r outside.
D) V increases linearly inside the sphere and decreases as 1/r outside.

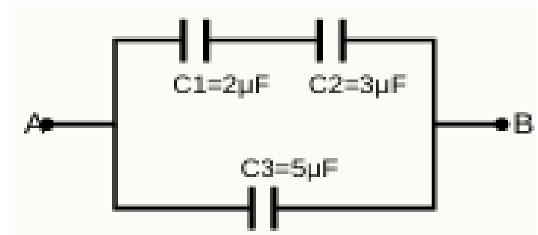
40. A parallel plate capacitor with air between the plates has an initial energy density u_0. It is connected to a battery of steady voltage V. While still connected to the battery, a dielectric slab of dielectric constant K = 2 is introduced to completely fill the space between the plates. What will be the new energy density and the ratio of the new charge to the old charge ($Q_{new} : Q_{old}$)?

A) $2u_0$ and 1:2 B) $4u_0$ and 2:1 C) $2u_0$ and 2:1 D) $u_0/2$ and 2:1

41. Based on the capacitor network shown in the figure, calculate the equivalent capacitance between terminals A and B.

(Given: $C_1 = 2\ \mu F$, $C_2 = 3\ \mu F$, $C_3 = 5\ \mu F$)

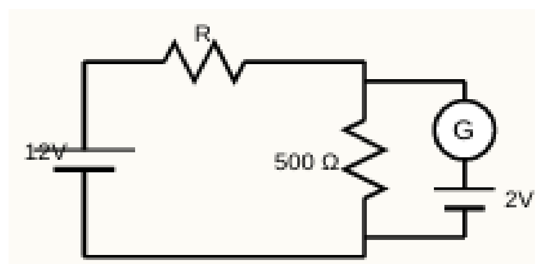
A) 10 µF
B) 6.2 µF
C) 1.2 µF
D) 5 µF



42. Two electric bulbs rated as 25W, 220V and 100W, 220V are connected in series across a 440V power supply. Which of the following statements is true?

A) Only the 25W bulb will fuse. B) Only the 100W bulb will fuse.
C) Both bulbs will fuse. D) Neither bulb will fuse.

43. In the given circuit, the galvanometer G shows zero deflection. If both batteries have negligible internal resistance, what is the value of the unknown resistor R? (Given: Main battery = 12V, parallel resistor = 500 Ω, galvanometer branch battery = 2V)



A) 2500 Ω B) 2000 Ω
C) 1000 Ω D) 500 Ω

44. For a given thermocouple, the neutral temperature is 270°C and the temperature of the cold junction is 20°C. What is the temperature of inversion?

A) 290°C B) 520°C C) 540°C D) 250°C

45. An alternating current circuit has an inductor of 0.2 H, a capacitor of 20 µF, and a resistor of 10 Ω connected in series. The Q-factor (quality factor) of this circuit is:

A) 5 B) 10 C) 20 D) 50

46. The magnetic susceptibility of a paramagnetic substance is x at 27°C. At what temperature will its susceptibility become x/3?

A) 9°C B) 81°C C) 627°C D) 900°C

47. In the Hall effect experiment, a current-carrying conductor is placed in a uniform magnetic field. The generated Hall voltage is directly proportional to:

A) The thickness of the conductor B) The drift velocity of the charge carriers
C) The inverse of the magnetic field strength D) The charge carrier density

48. An electron and a proton enter a uniform magnetic field with the exact same kinetic energy. If they enter perpendicularly to the field lines, what is the ratio of the radii of their circular paths ($r_e : r_p$)? (Assume m_e and m_p are their respective masses).

A) 1:1 B) $\sqrt{m_e} : \sqrt{m_p}$ C) $m_e : m_p$ D) $\sqrt{m_p} : \sqrt{m_e}$

49. A metallic rod of length L is rotated with an angular velocity ω about an axis passing through one of its ends and perfectly perpendicular to a uniform magnetic field B. What is the magnitude of the induced emf across the ends of the rod, and which end is at a higher electric potential if the magnetic field is directed into the page and the rotation is clockwise?

A) $\frac{1}{2} BL^2$, pivoted end is positive B) $B\omega L^2$, free end is positive
C) $\frac{1}{2} B\omega L^2$, free end is positive D) $B\omega L^2$, pivoted end is positive

50. To minimize the power loss due to the formation of eddy currents in an alternating current transformer, the core is typically:

A) Made of a solid block of highly conductive copper. B) Made of soft iron specifically to increase hysteresis loss.
C) Laminated with thin sheets of steel insulated from each other. D) Made of a completely non-magnetic material.

CHEMISTRY

51. An unknown organic compound contains C, H, and O. Complete combustion of 0.44 g of the compound yields 0.88 g of CO_2 and 0.36 g of H_2O. If the vapor density of the compound is 44, what is its actual molecular formula?
A) C_2H_4O B) $C_3H_6O_2$ C) $C_4H_8O_2$ D) $C_4H_{10}O$

52. 20.0 g of calcium carbonate ($CaCO_3$) is allowed to react with 200 mL of 1.0 M HCl. The volume of CO_2 gas collected is found to be 1.792 L at STP. What is the percentage yield of the reaction? (Molar mass: $CaCO_3$ = 100 g/mol, Molar volume at STP = 22.4 L/mol)
A) 75% B) 80% C) 90% D) 100%

53. Which of the following sets of quantum numbers describes an electron that requires the highest ionization energy to be removed from an isolated gaseous multi-electron atom?
A) n = 3, l = 2, m = 0, s = +1/2 B) n = 4, l = 0, m = 0, s = -1/2
C) n = 3, l = 1, m = -1, s = +1/2 D) n = 4, l = 1, m = 0, s = +1/2

54. Consider the isoelectronic species N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+}, and Al^{3+}. Which of the following statements correctly identifies the trend in their ionic radii and the fundamental reason for this trend?
A) Radii increase from N^{3-} to Al^{3+} due to increasing effective nuclear charge.
B) Radii decrease from N^{3-} to Al^{3+} due to an increasing effective nuclear charge pulling the same number of electrons closer.
C) Radii decrease from N^{3-} to Al^{3+} because cations naturally have fewer shells than anions.
D) Radii remain constant as they all contain exactly 10 electrons.

55. The molecules BF_3, NH_3, and ClF_3 all contain a central atom bonded to three surrounding atoms. Based on VSEPR theory and hybridization, what are their respective molecular geometries and central atom hybridizations?
A) All are trigonal planar, sp^2
B) Trigonal planar (sp^2), Trigonal pyramidal (sp^3), T-shaped (sp^3d)
C) Trigonal planar (sp^2), Trigonal pyramidal (sp^3), Trigonal planar (sp^2)
D) T-shaped (sp^3d), Trigonal pyramidal (sp^3), T-shaped (sp^3d)

56. During a quantitative redox titration in an acidic medium, Mohr's salt [$FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$] acts as a reducing agent against $KMnO_4$. What is the equivalent weight of Mohr's salt in this specific reaction? (Given: Molar mass of Mohr's salt = 392 g/mol)
A) 392 B) 196 C) 78.4 D) 39.2

57. Two real gases, A and B, have critical temperatures of 304 K and 126 K respectively. Which of the following deductions is chemically accurate regarding their deviation from ideal gas behavior and ease of liquefaction?
A) Gas B deviates more from ideal behavior and is easier to liquefy.
B) Gas A has stronger intermolecular forces, deviates more from ideality, and is easier to liquefy.
C) Gas A acts more like an ideal gas at room temperature than Gas B.
D) Critical temperature has no correlation with van der Waals forces or ideal gas deviation.

58. An ionic compound crystallizes in a unit cell where atoms of type A occupy the corners of a cube, atoms of type B occupy the centers of all faces, and atoms of type C occupy one fourth of the available tetrahedral voids. What is the empirical formula of the compound?
A) AB_3C_2 B) $A_8B_6C_2$ C) AB_3C_4 D) A_4B_4C

59. Consider the industrial synthesis of ammonia: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, where $\Delta H = -92.4$ kJ/mol. What will be the specific effect of adding an inert gas (e.g., Argon) to the system while keeping the pressure constant?
A) The equilibrium will shift towards the products (forward reaction).
B) The equilibrium will shift towards the reactants (backward reaction).
C) There will be no shift in the equilibrium position.
D) The equilibrium constant Kc will inherently decrease.

60. A 50 mL sample of H_2SO_4 requires 40 mL of 0.1 N NaOH for complete neutralization. If this exact same acid is utilized to neutralize 20 mL of a 0.2 N Na_2CO_3 solution, what specific volume of the acid is required?
A) 25 mL B) 50 mL C) 40 mL D) 100 mL

61. What is the pH of a solution obtained by mixing 100 mL of 0.1 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) and 50 mL of 0.1 M NaOH? (Given: log 1.8 ≈ 0.26)
A) 2.37 B) 4.74 C) 5.00 D) 9.26

62. The solubility product (K_{sp}) of Barium Sulfate ($BaSO_4$) is 1.0×10^{-10}. Taking the common ion effect into account, what is its molar solubility in a 0.1 M solution of Na_2SO_4?
A) 1.0×10^{-5} M B) 1.0×10^{-9} M C) 1.0×10^{-10} M D) 1.0×10^{-11} M

63. For a certain chemical reaction A → B, the half-life is 50 minutes when the initial concentration of A is 0.1 M, and the half-life drastically becomes 25 minutes when the initial concentration is increased to 0.2 M. What is the overall order of this reaction?
A) Zero order B) First order C) Second order D) Third order

64. A heterogeneous catalyst lowers the activation energy of a specific chemical reaction by exactly 10 kJ/mol at an operating temperature of 300 K. Assuming the Arrhenius frequency factor (A) remains constant, by approximately what factor does the rate of the forward reaction increase? (Use $R \approx 8.314$ J K^{-1} mol^{-1}, and $e^4 \approx 54.6$)
A) 4 times B) 10 times C) 55 times D) 100 times

65. A standard galvanic cell is constructed consisting of a Cu/Cu^{2+} half-cell and an Ag/Ag^+ half-cell. Given $E°(Cu^{2+}/Cu) = +0.34$ V and $E°(Ag^+/Ag) = +0.80$ V. If the concentration of Cu^{2+} is modified to 1.0 M and Ag^+ is dropped to 0.1 M, what is the new cell potential (E_{cell}) at 298 K? (Use 2.303RT/F = 0.0591)
A) 0.460 V B) 0.401 V C) 0.342 V D) 0.519 V

66. The standard enthalpies of combustion at 298 K for methane gas (CH_4), graphite (C), and dihydrogen (H_2) are -890.3 kJ/mol, -393.5 kJ/mol, and -285.8 kJ/mol respectively. Utilizing Hess's Law, what is the standard enthalpy of formation of methane?
A) -74.8 kJ/mol B) +74.8 kJ/mol C) -157.0 kJ/mol D) -1569.6 kJ/mol

67. A piece of wood excavated from an ancient artifact has a Carbon-14 radioactive activity of 4.0 disintegrations per minute per gram (dpm/g) of carbon. The activity of C-14 in modern living wood is known to be 16.0 dpm/g. If the half-life of Carbon-14 is 5730 years, what is the approximate archaeological age of the artifact?
A) 2865 years B) 5730 years C) 11460 years D) 17190 years

68. Which intermediate is primarily formed by the homolytic fission of a covalent bond?
A) Carbocation B) Carbanion C) Free radical D) Carbene

69. The acidic nature of terminal alkynes is attributed to which hybridization state of carbon?
A) sp^3 B) sp^2 C) sp D) dsp^2

70. Which of the following alkyl halides is most reactive towards an SN_1 mechanism?
A) CH_3Cl B) CH_3CH_2Cl C) $(CH_3)_2CHCl$ D) $(CH_3)_3CCl$

71. The industrial fermentation of sugar (sucrose) primarily produces which type of alcohol?
A) Methanol B) Ethanol C) Propanol D) Butanol

72. Which compound yields a secondary alcohol upon reaction with methylmagnesium bromide followed by hydrolysis?
A) Formaldehyde B) Acetaldehyde C) Acetone D) Ethylene oxide

73. Williamson's synthesis is unsuitable for preparing di-tert-butyl ether because:
A) Tert-butoxide is a weak base
B) Tert-butyl halide undergoes elimination
C) The reaction is highly reversible
D) Steric hindrance prevents hemolysis

74. To separate a mixture of primary, secondary, and tertiary amines using Hoffmann's method, the specific reagent used is:
A) Hinsberg reagent B) Diethyl oxalate C) Phosgene D) Nitrous acid

75. An organocopper compound (Gilman reagent, R_2CuLi) reacts with acid chlorides to predominantly yield:
A) Aldehydes B) Ketones C) 3° Alcohols D) Esters

76. The relative reactivity of carboxylic acid derivatives towards nucleophilic acyl substitution is:

A) Ester > Amide > Anhydride > Chloride
B) Chloride > Anhydride > Ester > Amide
C) Anhydride > Chloride > Amide > Ester
D) Amide > Ester > Chloride > Anhydride

77. Which effect explains why phenol is significantly more acidic than aliphatic alcohols?
A) Inductive effect (+I)
B) Resonance effect (-R)
C) Electromeric effect
D) Resonance stabilization of phenoxide

78. Hydroboration-oxidation of propene yields:
A) Propan-1-ol B) Propan-2-ol C) Propane D) Propanal

79. In the context of gasoline quality, an octane number of 100 is assigned to:
A) n-Heptane B) Iso-octane C) Benzene D) Cetane

80. Chlorobenzene is notably less reactive towards nucleophilic substitution than ethyl chloride due to:
A) Sp3 hybridization of the C-Cl carbon
B) Partial double bond character of the C-Cl bond
C) The +I effect of the benzene ring
D) Instability of the phenyl radical

81. Reduction of nitrobenzene in a neutral medium (e.g, using Zn dust and NH_4Cl) yields:
A) Aniline B) N-Phenylhydroxylamine C) Azobenzene D) Nitrosobenzene

82. The major product of adding HBr to 2-methylpropene in the presence of an organic peroxide is:
A) 2-Bromo-2-methylpropane
B) 1-Bromo-2-methylpropane
C) 2-Bromo-1-methylpropane
D) 1-Bromo-2-methylpropene

83. Benzaldehyde undergoes the Cannizzaro reaction strictly because it:
A) Has a reactive alpha-hydrogen
B) Lacks an alpha-hydrogen
C) Is an aromatic aldehyde
D) Forms a highly stable enolate

84. The reaction of a Grignard reagent (RMgX) with solid carbon dioxide (dry ice) followed by acid hydrolysis produces:
A) An Aldehyde B) A Ketone C) A Carboxylic acid D) An Ester

85. Which of the following hydrogen halides possesses the highest boiling point, and which one acts as the strongest reducing agent?
A) HI and HF B) HF and HI C) HCl and HI D) HF and HBr

86. In the extraction of iron, limestone is added to the blast furnace. What is its exact metallurgical role?
A) It acts as a reducing agent.
B) It acts as an acidic flux to remove basic impurities.
C) It decomposes to form a basic flux that removes acidic gangue as slag.
D) It lowers the melting point of the iron ore

87. Which of the following nitrogen oxides is highly reactive and acts as a major catalyst in the depletion of the stratospheric ozone layer?
A) N_2O (Nitrous oxide)
B) NO_2 (Nitrogen dioxide)
C) N_2O_5 (Dinitrogen pentoxide)
D) NO (Nitric oxide)

88. According to Crystal Field Theory, what is the crystal field stabilization energy (CFSE) for a high-spin d_4 octahedral complex?
A) $-0.6\ \Delta o$ B) $-1.6\ \Delta o$ C) $-0.4\ \Delta o$ D) $0.0\ \Delta o$

89. The biological condition known as Itai-Itai disease is caused by chronic toxicity of which heavy metal?
A) Mercury (Hg) B) Lead (Pb) C) Cadmium (Cd) D) Arsenic (As)

90. In the MacArthur-Forrest cyanide process for the extraction of silver, what is the specific role of atmospheric oxygen?
A) It reduces silver ions to metallic silver.
B) It acts as an oxidizing agent to convert Ag to Ag^+ so it can form a soluble complex.
C) It oxidizes cyanide ions to cyanate to prevent toxicity.
D) It roasts the silver ore prior to complexation.

91. What sequence of visual changes occurs when an excess of stannous chloride ($SnCl_2$) is gradually added to an aqueous solution of corrosive sublimate ($HgCl_2$)?
A) A grey precipitate forms immediately.
B) A white precipitate forms which dissolves in excess SnCl2.
C) A white precipitate forms which eventually turns grey.
D) A black precipitate forms which turns white.

92. Which of the following metals are commercially purified using the 'poling' method to remove their respective metal oxide impurities?
A) Iron and Zinc B) Copper and Tin C) Mercury and Silver D) Aluminum and Magnesium

93. When H_2S gas is passed through an acidified solution of potassium dichromate ($K_2Cr_2O_7$), the solution turns green. What is the chemical reason for this change?

A) Formation of $Cr_2(SO_4)_3$, while H_2S acts as a reducing agent.

B) Formation of CrO_4(2-), while H_2S acts as an oxidizing agent.

C) Precipitation of green colloidal sulphur.

sD) Formation of a complex thio-chromate ion.

94. Which statement accurately describes the stoichiometry of the biological sodium potassium pump per molecule of ATP hydrolyzed?

A) 2 Na^+ are pumped out of the cell, 3 K^+ are pumped in. B) 3 Na^+ are pumped out of the cell, 2 K^+ are pumped in.

C) 3 Na^+ and 3 K^+ are both pumped out of the cell. D) 2 Na^+ are pumped into the cell, 3 K^+ are pumped out.

95. In the modern Contact Process for manufacturing sulphuric acid, what is the primary catalyst used for the oxidation of sulfur dioxide to sulfur trioxide?

A) Platinized asbestos B) Vanadium(V) oxide (V_2O_5) C) Finely divided iron D) Nitric oxide

96. Which of the following is an example of a condensation polymer used widely in the textile industry?

A) Teflon B) Polyvinyl chloride (PVC) C) Nylon-6,6 D) Polystyrene

97. Which of the following sequences of reagents is most appropriate for converting phenol into aspirin (acetylsalicylic acid)?

A) NaOH/ CO_2 followed by H^+ , then CH_3COCl B) $CHCl_3$/NAOH followed by H^+ , then $(CH_3CO)_2O$

C) Zn dust followed by CH_3Cl/ $AlCl_3$, then $KMnO_4$ D) $NaNO_2$ / HCl followed by CuCN, then H_3O^+

98. Which separation technique is most suitable for purifying a liquid that decomposes at or below its normal boiling point?

A) Simple distillation B) Fractional distillation C) Vacuum distillation D) Fractional crystallization

99. In Lassaigne's test for the detection of nitrogen in an organic compound, the appearance of a characteristic Prussian blue color is due to the formation of which complex compound?

A) $Na_4[Fe(CN)_6]$ B) $Fe_4[Fe(CN)_6]_3$ C) $Fe_3[Fe(CN)_6]_2$ D) $Na_2[Fe(CN)_5NO]$

100. In an iodometric titration, a standard solution of sodium thiosulfate ($Na_2S_2O_3$) is used to titrate the liberated iodine. What is the equivalent weight of sodium thiosulfate in this reaction? (Where M = molar mass of $Na_2S_2O_3$).

A) M B) M/2 C) M/3 D) 2M

BOTANY

101. Which of the following describes the behavior of a competitive enzyme inhibitor?

A) It decreases both V_{max} and Km. B) It decreases V_{max} but leaves Km unchanged.

C) It increases Km but leaves Vmax unchanged. D) It binds to an allosteric site on the enzyme.

102. Cellulose is highly stable and difficult to digest for most animals because of its specific structure. Which linkage is characteristic of cellulose?

A) α-1,4-glycosidic linkages between glucose units B) β-1,4-glycosidic linkages between glucose units

C) α-1,6-glycosidic linkages between glucose units D) β-1,2-glycosidic linkages between fructose and glucose

103. In the 5-Kingdom classification system proposed by R.H. Whittaker, what is the primary basis for placing organisms into the kingdom Monera?

A) Presence of a rigid cellulosic cell wall B) Absence of a defined nucleus and membrane-bound organelles

C) Multicellular body organization and heterotrophic nutrition D) Ability to perform oxygenic photosynthesis

104. Ophiocordyceps sinensis (Yarsagumba), a highly valued medicinal organism in Nepal, belongs to which specific group of fungi?

A) Phycomycetes B) Ascomycetes C) Basidiomycetes D) Deuteromycetes

105. Rauwolfia serpentina, commonly known as Sarpagandha, is primarily used in traditional and modern medicine for the treatment of:

A) A. Malaria B) Hypertension (high blood pressure) C) Diabetes mellitus D) Tuberculosis

106. Which of the following floral characteristics is a diagnostic feature of the family Brassicaceae?

A) Epipetalous stamens and swollen axile placenta B) Tetradynamous stamens and a false septum (replum)

C) Diadelphous stamens and marginal placentation D) Trimerous flowers and basal placentation

107. In the life cycle of the pteridophyte Dryopteris, the prothallus represents the:

A) Diploid sporophytic phase
B) Independent, haploid gametophytic phase
C) Parasitic gametophytic phase dependent on the sporophyte
D) Spore-producing organ

108. What is the ploidy level of the endosperm in Pinus, and when is it formed?
A) Diploid (2n), formed before fertilization
B) Triploid (3n), formed after double fertilization
C) Haploid (n), formed before fertilization
D) Haploid (n), formed after fertilization

109. Which of the following is the reserve food material in the red algae (*Rhodophyceae*)?
A) Starch
B) Floridean starch
C) Laminarin and mannitol
D) Glycogen

110. Heterocysts found in some filamentous Cyanobacteria like Nostoc and Anabaena are specialized primarily for:
A) Photosynthesis under high light intensity
B) Asexual reproduction
C) Nitrogen fixation
D) Storage of lipid granules

111. Viruses are considered obligate intracellular parasites. What is the chemical composition of a typical plant virus?
A) DNA and a lipid envelope
B) RNA and a protein coat
C) Only naked RNA without a protein coat
D) DNA and complex carbohydrates

112. During the nitrogen cycle, the biological conversion of ammonia (NH_3) to nitrite (NO^{2-}) and then to nitrate (NO^{3-}) is carried out by specific chemoautotrophic bacteria. Which bacteria is responsible for the conversion of nitrite to nitrate?
A) Nitrosomonas
B) Rhizobium
C) Nitrobacter
D) Pseudomonas

113. In a hydrosere (ecological succession in a water body), what represents the pioneer community?
A) Submerged aquatic plants
B) Floating aquatic plants
C) Phytoplankton
D) Marsh meadow vegetation

114. Acid rain is a severe ecological imbalance primarily caused by the emission of which gases into the atmosphere?
A) Carbon dioxide and Carbon monoxide
B) Chlorofluorocarbons and Methane
C) Sulfur dioxide and Nitrogen oxides
D) Ozone and Peroxyacetyl nitrate

115. Which of the following biotic interactions represents an 'Amensalism' relationship?
A) One species benefits while the other is unaffected (+ / 0)
B) Both species benefit (+ / +)
C) One species is harmed while the other is unaffected (- / 0)
D) One species benefits at the expense of the other (+ / -)

116. Which cell organelle is characterized by the presence of hydrolytic enzymes that function optimally at an acidic pH?
A) Peroxisome
B) Lysosome
C) Golgi apparatus
D) Endoplasmic reticulum

117. During the cell cycle, in which phase does the amount of DNA double without an increase in the number of chromosomes?
A) G1 phase
B) S phase
C) G_2 phase
D) M phase

118. During prophase I of meiosis, the physical exchange of genetic material between non-sister chromatids of homologous chromosomes (crossing over) occurs in which specific substage?
A) Leptotene
B) Zygotene
C) Pachytene
D) Diplotene

119. According to the widely accepted Fluid Mosaic Model proposed by Singer and Nicolson, the cell membrane is primarily composed of:
A) A rigid bilayer of proteins embedded with lipids
B) A fluid bilayer of phospholipids with embedded and peripheral proteins
C) A single layer of lipids coated by proteins on both sides
D) Cellulose fibers interlinked by pectin

120. Which of the following statements provides the strongest support for the endosymbiotic theory regarding mitochondria and chloroplasts?
A) They are part of the endomembrane system connected to the ER.
B) They lack ribosomes and rely entirely on the nucleus for protein synthesis.
C) They contain circular DNA and 70S ribosomes, similar to prokaryotes.
D) They reproduce exclusively through complete de novo synthesis during mitosis.

121. In the context of the Central Dogma of molecular biology, what is the role of transfer RNA (tRNA)?
A) To carry the genetic message from DNA to the ribosome
B) To act as a structural component of the ribosome
C) To carry specific amino acids to the ribosome during translation
D) To synthesize mRNA during transcription

122. In a dihybrid test cross involving linked genes, the resulting phenotypic ratio heavily deviates from the Mendelian 1:1:1:1 ratio. This deviation, showing a higher frequency of parental types, is due to:
A) Incomplete dominance B) Independent assortment C) Incomplete linkage D) Epistasis
123. A color-blind man marries a woman with normal vision whose father was color-blind. What is the probability that their sons will be color-blind?
A) 0% B) 25% C) 50% D) 100%
124. Which type of chromosomal mutation involves the loss of a single chromosome from a diploid set, represented as (2n - 1)?
A) Nullisomy B) Monosomy C) Trisomy D) Tetrasomy
125. Klinefelter's syndrome is a genetic disorder caused by aneuploidy. What is the characteristic karyotype of an individual with this syndrome?
A) 45, XO B) 47, XXY C) 47, XYY D) 47, +21 (Trisomy 21)
126. According to Chargaff's rule, if a double-stranded DNA molecule contains 20% adenine, what is the percentage of cytosine in the DNA?
A) 20% B) 30% C) 40% D) 60%
127. Which of the following plant tissues is composed of living cells with localized pectin thickenings at the corners, providing mechanical support to growing parts of the plant?
A) Parenchyma B) Collenchyma C) Sclerenchyma D) Xylem vessels
128. In a transverse section (T.S.) of a typical monocot stem, what is the characteristic arrangement and nature of the vascular bundles?
A) Arranged in a ring, collateral and open B) Scattered, collateral, and closed with a bundle sheath
C) Arranged in a ring, bicollateral and open D) Scattered, radial and closed
129. Casparian strips, which regulate water flow into the vascular cylinder by blocking the apoplastic pathway, are found in the:
A) Epidermis of the stem B) Pericycle of the root C) Endodermis of the root D) Bundle sheath of the leaf
130. A plant cell has an Osmotic Pressure (OP) of 15 atm and a Turgor Pressure (TP) of 5 atm. What is the Diffusion Pressure Deficit (DPD) of this cell, and what will happen if it is placed in pure water?
A) DPD is 20 atm; water will leave the cell. B) DPD is 10 atm; water will enter the cell.
C) DPD is 15 atm; no net movement of water. D) DPD is 5 atm; the cell will plasmolyze.
131. In the C_4 pathway (Hatch-Slack pathway) of photosynthesis, what is the primary carbon dioxide acceptor molecule?
A) Ribulose-1,5-bisphosphate (RuBP) B) Phosphoenolpyruvate (PEP)
C) Oxaloacetic acid (OAA) D) Phosphoglyceric acid (PGA)
132. The active transport mechanism widely accepted for the opening and closing of stomata relies primarily on the influx and efflux of which ion in the guard cells?
A) Sodium (Na^+) B) Calcium (Ca^{2+}) C) Potassium (K^+) D) Magnesium (Mg^{2+})
133. During non-cyclic photophosphorylation (Z-scheme) in the light-dependent reactions of photosynthesis, what are the terminal products generated?
A) ATP only B) ATP and O_2 only C) ATP, NADPH, and O_2 D) NADPH and O_2 only
134. In aerobic respiration, the Krebs cycle (TCA cycle) takes place in the:
A) Cytoplasm B) Inner mitochondrial membrane C) Mitochondrial matrix D) Chloroplast stroma
135. Which plant growth regulator is famously responsible for promoting internode elongation in rosette plants prior to flowering, a process known as 'bolting'?
A) Auxin B) Gibberellin C) Cytokinin D) Abscisic acid
136. In angiosperms, double fertilization is a unique event. What are the two distinct fusion products of this process?
A) Two diploid zygotes B) A diploid zygote and a triploid primary endosperm nucleus (PEN)
C) A haploid embryo and a diploid endosperm D) A triploid zygote and a diploid seed coat
137. If a typical dicot leaf cell contains 24 chromosomes, what will be the chromosome number in the synergid cells and the primary endosperm nucleus (PEN) respectively?
A) 12 and 36 B) 24 and 36 C) 12 and 24 D) 24 and 48

138. In plant tissue culture, what term describes the unorganized, undifferentiated mass of dividing cells produced from an explant?
A) Embryoid B) Callus C) Protoplast D) Meristem

139. Which bacterium is widely recognized as a 'natural genetic engineer' of plants because of its ability to transfer a segment of its Ti plasmid (T-DNA) into the plant genome?
A) Bacillus thuringiensis B) Agrobacterium tumefaciens
C) Rhizobium leguminosarum D) Escherichia coli

140. The water fern Azolla is extensively used as a biofertilizer in rice fields because it harbors a symbiotic nitrogen-fixing cyanobacterium known as:
A) Nostoc B) Anabaena azollae C) Spirulina D) Oscillatoria

ZOOLOGY

141. According to the Oparin-Haldane theory of the origin of life, the primitive atmosphere of the Earth was highly reducing because it lacked which of the following free gases?
A) Hydrogen B) Oxygen C) Methane D) Ammonia

142. In the Miller-Urey experiment, an electric spark was passed through a specific mixture of gases to simulate early Earth conditions. What primary organic compounds were observed in the resulting liquid?
A) Nucleic acids B) Amino acids C) Complex carbohydrates D) Lipids

143. Tracing human evolution from Ramapithecus to modern man, which of the following ancestors is widely recognized as the first to exhibit a fully upright posture and bipedal locomotion?
A) Ramapithecus B) Australopithecus C) Homo habilis D) Neanderthal man

144. The primary diagnostic feature used to classify the phylum Protozoa into its respective classes is the:
A) Mode of nutrition B) Type of locomotory organelles C) Number of nuclei D) Method of reproduction

145. Which of the following is an exclusive, defining diagnostic feature of the phylum Chordata that must be present at some stage of their life cycle?
A) Ventral solid nerve cord B) Presence of a notochord C) Jointed appendages D) Pseudocoelom

146. A diagnostic feature of mammals, distinguishing them from all other chordates, is the presence of:
A) four-chambered heart B) Internal fertilization C) Mammary glands and hair D) Lungs for respiration

147. To which phylum does an animal belong if it possesses a soft body, a muscular foot, and a visceral mass usually covered by a calcareous shell?
A) Arthropoda B) Echinodermata C) Mollusca D) Annelida

148. Which type of epithelial tissue lines the inner surface of the stomach and intestine, playing a major role in secretion and absorption?
A) Squamous epithelium B) Cuboidal epithelium C) Columnar epithelium D) Stratified epithelium

149. Tendons and ligaments are highly specialized forms of which type of tissue?
A) Muscular tissue B) Nervous tissue C) Dense connective tissue D) Areolar connective tissue

150. A histology student observes muscle tissue that is striated but involuntary, featuring intercalated discs. This specific tissue is located in the:
A) Biceps B) Stomach wall C) Heart D) Urinary bladder

151. In nervous tissue, the structural and functional unit responsible for receiving and transmitting nerve impulses is the:
A) Neuron B) Neuroglia C) Myelin sheath D) Dendrit

152. In the life cycle of Plasmodium, the infective stage that is introduced into the human body by the bite of an infected female Anopheles mosquito is the:
A) Trophozoite B) Sporozoite C) Merozoite D) Gametocyte

153. Which species of Plasmodium is responsible for causing malignant tertian malaria, the most severe and fatal type of malaria?
A) *Plasmodium vivax* B) *Plasmodium ovale* C) *Plasmodium malariae* D) *Plasmodium falciparum*

154. In the earthworm (Pheretima), which segments are covered by the prominent, dark glandular band known as the clitellum?

A) Segments 10, 11, 12 B) Segments 14, 15, 16 C) Segments 18, 19, 20 D) Segments 24, 25, 26

155. The earthworm is of great economic importance to agriculture primarily because it:

A) Acts as a biocontrol agent against crop pests
B) Improves soil fertility and aeration through burrowing and casting
C) Pollinates underground crops
D) Fixes atmospheric nitrogen in the soil

156. During hibernation, a frog (Rana) relies entirely on which method for its respiratory gas exchange?

A) Pulmonary respiration (lungs) B) Buccal respiration
C) Cutaneous respiration (skin) D) Branchial respiration (gills)

157. Structurally, the heart of a frog differs from a human heart because it consists of:

A) Two atria and two ventricles B) Two atria and one ventricle
C) One atrium and two ventricles D) One atrium and one ventricle

158. If a patient's liver is severely damaged, the digestion of which of the following components will be most immediately impaired due to a lack of emulsification?

A) Carbohydrates B) Proteins C) Fats D) Nucleic acids

159. In the human digestive system, the highly acidic food leaving the stomach (chyme) is neutralized in the duodenum primarily by secretions from the:

A) Salivary glands B) Pancreas and Liver C) Gastric glands D) Large intestine

160. In the transport of respiratory gases, carbon dioxide is transported from tissues to the lungs primarily in the form of:

A) Dissolved gas in plasma B) Carboxyhemoglobin
C) Bicarbonate ions in plasma D) Carbonic acid inside RBCs

161. Which respiratory disorder is characterized by the destruction of alveolar walls, decreasing the surface area for gas exchange, and is most often caused by cigarette smoking?

A) Asthma B) Pneumonia C) Emphysema D) Tuberculosis

162. The term "Cardiac Output" refers to the volume of blood:

A) Present in the heart at rest B) Pumped out by each ventricle per minute
C) Pumped out by each ventricle per beat D) Entering the atria per minute

163. A person with blood group AB is considered a universal recipient because their blood:

A) Has both Anti-A and Anti-B antibodies in the plasma B) Lacks both Anti-A and Anti-B antibodies in the plasma
C) Lacks both A and B antigens on the red blood cells D) Has no Rh factor

164. In the excretory system, the physiological process of ultrafiltration for urine formation occurs specifically in the:

A) Loop of Henle B) Proximal Convoluted Tubule C) Bowman's capsule and Glomerulus D) Collecting duct

165. A patient's urine test reveals a high concentration of glucose (Glycosuria). This renal disorder is most commonly associated with a malfunction of the:

A) Liver B) Endocrine pancreas (Diabetes mellitus)
C) Adrenal glands D) Posterior pituitary (Diabetes insipidus)

166. In the Central Nervous System (CNS), which part of the brain acts as the primary relay center for sensory and motor impulses?

A) Cerebellum B) Thalamus C) Medulla oblongata D) Corpus callosum

167. During the conduction of a nerve impulse, the rapid influx of which ion is responsible for the depolarization of the axonal membrane?

A) Potassium ($K^{+)}$) B) Calcium (Ca^{++}) C) Sodium (Na^{+}) D) Chloride (Cl^{-})

168. Which structure in the human eye functions analogously to the diaphragm of a camera by regulating the amount of light that enters?

A) Cornea B) Lens C) Iris D) Retina

169. The endocrine gland responsible for regulating the body's basal metabolic rate (BMR) through the secretion of thyroxine is the:

A) Thyroid gland B) Adrenal gland C) Parathyroid gland D) Thymus

170. Goiter is an endocrine disorder caused by the enlargement of the thyroid gland, typically due to a dietary deficiency of:

A) Calcium B) Iron C) Iodine D) Sodium

171. In the human male reproductive system, the process of spermatogenesis occurs directly within the:

A) Epididymis B) Seminiferous tubules C) Prostate gland D) Vas deferens

172. During the menstrual cycle, the sudden surge of which hormone directly triggers ovulation?

A) Estrogen B) Progesterone C) Luteinizing Hormone (LH) D) Follicle Stimulating Hormone (FSH)

173. Which of the following diseases is caused by the bacterium Mycobacterium tuberculosis?

A) Typhoid B) TB (Tuberculosis) C) Cholera D) Influenza

174. Candidiasis is an infection commonly seen in immunocompromised individuals. Its causative agent is a:

A) Virus B) Bacterium C) Fungus D) Protozoan

175. Immunity acquired by an individual after recovering from a natural infection, such as chickenpox, is classified as:

A) Innate immunity B) Artificial active immunity C) Natural active acquired immunity D) Passive immunity

176. The Polio Sabin vaccine, administered orally as drops, is an example of which type of vaccine?

A) Inactivated vaccine B) Live attenuated vaccine C) Toxoid vaccine D) Recombinant vaccine

177. Amniocentesis is a medical technology procedure primarily used for:

A) In-Vitro Fertilization B) Detecting genetic and chromosomal abnormalities in a fetus

C) Cloning transgenic animals D) Treating microbial infections in the uterus

178. In applied microbiology for sewage treatment, the "biological treatment" or secondary treatment relies on the action of:

A) Chemical coagulants B) Bio-control agents like ladybugs

C) Heterotrophic microbes (bacteria and fungi) D) Dairy microbes like Lactobacillus

179. The stereotypic, directional movement of an animal toward or away from a specific stimulus (like a moth flying toward light) is termed:

A) Reflex action B) Migration C) Taxis D) Adaptation

180. In conservation biology, regions that harbor a very high level of species richness and a high degree of endemism, but are severely threatened by habitat destruction, are designated as:

A) Ramsar sites B) IUCN categories C) Protected areas D) Biodiversity Hotspot1

MAT

181. PNEUMONIA : LUNGS, GLAUCOMA : ?

A) Bones B) Eyes C) Liver D) Kidneys

182. Identify the word choice from the options below that acts as the absolute 'Odd One Out' in terms of its core semantic definition:

A) Disseminate B)Disperse C) Congregate D) Circulate

183. If a systemic encryption rule encodes the term 'CANDLE' into the string 'EDRIRL', which of the available choices represents the correct encrypted format for the word 'HEALTH' using the exact same operational rule?

A) JHEQZO B) JGDQZN C) KHEQZO D) JHEPAO

184. Pointing to a woman in a framed picture, Rohit noted: "Her mother's only biological daughter is my maternal grandmother's only daughter." If his maternal grandmother has no other children, what is Rohit's precise relationship to the woman in the photograph?

A) Brother B) Father C) Son D) Nephew

185. Find total numbers factors of 360.

A) 20 B) 24 C) 22 D) 26

186. Evaluate the mathematical progression governing the sequence and compute the value that must succeed the final item:

4, 9, 20, 43, 90, ?

A) 181 B)185 C) 189 D)195

187. An analytical testing lab increases its quarterly output targets by exactly 20% compounded annually. If the base output captured in the year 2024 was recorded as 5,000 baseline samples, what will be the calculated operational target volume for the year 2026?

A) 7,000 samples B) 7,200 samples C) 6,800 samples D) 7,500 samples

188. Find the missing value in the following table:

7	5	24
9	4	65
11	6	?

A) 55
B) 75
C) 85
D) 95

189. Exactly 3 medical systems analysts can fully calibrate 60 automated tracking lines inside an operational window of 4 days. Assuming a completely static, uniform performance rate across all personnel, how many days are required for 5 analysts to calibrate 150 matching tracking lines?

A) 5 days B) 6 days C) 7 days D) 8 days

190. Analyze the highly non-linear arithmetic series properties layout below and compute the correct value that must succeed the final item: 2, 3, 10, 39, 172, 885, ?

A) 5,310 B) 5,346 C) 5,286 D) 5,412

191. A logistics driver starts from the central sorting facility and travels 4 km directly North. He then executes a crisp 90-degree right turn and travels 3 km East. Following this, he executes another sharp right turn and moves forward for 4 km. How far and in which cardinal direction is the driver positioned relative to the coordinate origin of his path?

A) 3 Km West B) 3 Km East C) 4 Km North D) 7 Km South

192. Inside a linear queue of 35 corporate regional sales leads, Pradeep's location is recorded as precisely 12th when counted starting from the left boundary flank. What is his exact ranked index when enumerated starting directly from the right boundary flank?

A) 23rd B) 24th C) 22nd D) 25th

193. Five senior operations executives (A, B, C, D, and E) sit equidistantly around a round boardroom conference table, all facing directly inward toward the center point. The seating arrangement adheres to the following structural constraints: • Executive A sits immediately adjacent to Executive B.
• Executive C sits immediately adjacent to Executive D.
• Executive E sits directly to the immediate left of Executive D.
• Executive B is not positioned next to Executive C.
Based on these rules, which specific individual sits to the immediate right of Executive A?

A) Executive B B) Executive C C) Executive D D) Executive E

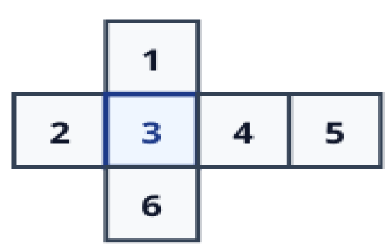
194. If the calendar day immediately following tomorrow is verified to be exactly two days prior to Friday, what specific day of the week corresponds to yesterday?

A) Sunday B) Monday C) Saturday D) Tuesday

195. Three flagship products (Product Alpha, Product Beta, and Product Gamma) are scheduled for market rollouts across three consecutive months (January, February, and March) managed respectively by three separate directors (Ramesh, Sita, and Hari), though not necessarily in that chronological order. The corporate planning parameters dictate the following:
1. Product Alpha was not launched in March, and was managed exclusively by Ramesh.
2. Sita successfully launched her assigned product in the month of February.
3. Product Gamma was not managed or launched by Sita. Identify the precise product managed by Director Hari and its corresponding launch month.

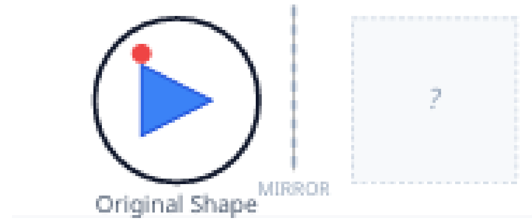
A) Product Alpha in January B) Product Gamma in March
C) Product Beta in February D) Product Gamma in February

196. Evaluate the structural configuration design of the 2D cube folding net presented below. If this flat 2D net layout is physically folded up into a perfect uniform 3D cube solid, which numbered face will lie directly opposite to the blue face marked '3'?



A) Face 1 B) Face 4
C) Face 5 D) Face 6

197. Identify the exact horizontal mirror reflection profile layout of the composite shape configuration shown below when a vertical reflecting line plane is set up directly on its right side:

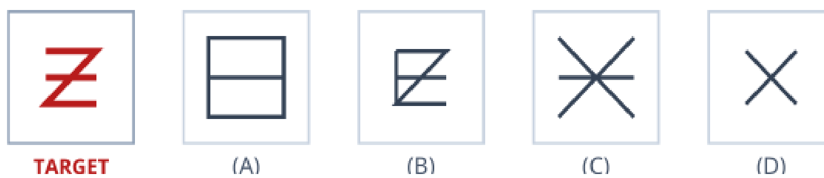


198. A flat square sheet of filter paper is sequentially folded twice: first precisely from right-to-left along its vertical line of symmetry, and secondly from top-to-bottom along its horizontal line of symmetry. A singular, clean circular hole punch is then executed entirely through the top-right corner coordinate of the final compressed layered square. Which structural pattern manifests across the sheet when it is completely unfolded back to its flat state?

A) Exactly four cut holes positioned near the outer corners of the paper sheet.
B) A single centered circular hole located exactly at the coordinate origin of the paper sheet.
C) Exactly two circular holes aligned horizontally along the structural midline.
D) Four separate circular holes forming a diamond pattern grouped near the midpoint

199. Embed Check: Determine which of the available option figures below houses the specific target line segment configuration embedded cleanly inside its structural geometry lines:

A) Figure 'A'
B) Figure 'B'
C) Figure 'C'
D) Figure 'D'



200. Abstract Sequence Rules Check: Analyze the synchronized three-part geometric transformation mapping across the progressive panels below, and determine the correct structure configuration for the missing Panel 4: